

An Investigation of an Important Family of Polynomial-Exponential Product Functions

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Abstract

This paper investigates a particular family of polynomial-exponential product functions whose n -th derivatives when evaluated at 0 are equal to $(k+n)^p$ where k, n, p are non-negative integers. It uses ordinary differentiation and algebraic manipulation to derive formulae for explicitly constructing functions of this family. I have independently discovered the family and its structure, without referencing existing mathematical literature.

Contents

1	Introduction	2
1.1	Notation	2
1.1.1	Polynomial-exponential product functions	2
1.1.2	The coefficient function	2
1.1.3	Summation variables	2
1.1.4	n -th derivatives	3
1.1.5	Equation numbering	3
1.2	Discovery of the family	3
1.2.1	Conception	3
1.2.2	Definition	5
2	Structure derivation	8
2.1	Structure of coefficients of polynomial-exponential product functions	8
2.2	General structure of the family	12

1 Introduction

1.1 Notation

1.1.1 Polynomial-exponential product functions

A polynomial-exponential product function is defined to be the product of a polynomial and an exponential function, both of the same variable. The general form for these functions is as follows:

$$f(x) = (\lambda_0 + \lambda_1 x + \cdots + \lambda_p x^p) e^{\alpha x}$$

Where:

- $\lambda_0, \lambda_1, \dots, \lambda_p$ are constant coefficients of terms in the polynomial factor,
- p is the non-negative integer degree of the polynomial factor,
- α is the exponential parameter.

Throughout this paper only polynomial-exponential product functions where $\alpha = 1$ are considered. Consequently, this exponential parameter may be suppressed from notation.

1.1.2 The coefficient function

Let's define the coefficient function

$$C(f, r)$$

to be the coefficient of the x^r term in the polynomial factor of the polynomial-exponential product function f . Where:

- f is a polynomial-exponential product function,
- r is a non-negative integer indicating the degree of the polynomial term whose coefficient is to be returned.

For example, $C(f, 2)$ would be equal to the coefficient of the x^2 term in the polynomial factor of f . Additionally, $C(f, 0)$ would be equal to the constant term in the polynomial factor (coefficient of x^0). $C(f, -1)$ does not exist, neither would $C(f, 0.5)$, as $r \geq 0$ and $r \in \mathbb{Z}$.

1.1.3 Summation variables

Throughout this paper, i will denote the primary index of summation. It should not be confused with the imaginary unit. For example:

$$\sum_{i=0}^n a_i$$

Where:

- i is the index of summation,
- 0 is the lower limit of summation,
- n is the upper limit of summation,
- a_i is the general term of the summation.

1.1.4 n -th derivatives

Throughout this paper I will use f^n to represent a function being differentiated n times. This is not to be confused with function composition. This may also appear in the format: f^{k+n} . This simply means differentiated $k + n$ times.

1.1.5 Equation numbering

Throughout this paper, I may number useful equations so that I can concisely reference these later on. These equations will not be reused in subsections different to those in which they were defined. Because of this, the equation numbers corresponding to particular equations may be reassigned. This should be taken into account as to not confuse these important equations with one another.

1.2 Discovery of the family

1.2.1 Conception

Let's begin with finding functions that satisfy $f^n(0) = n^q$, where q is a non-negative integer. Consider the following function:

$$f_1(x) = xe^x$$

This function constitutes as a polynomial-exponential product function whose polynomial factor has a degree of 1. Differentiating this function using the product rule gives us:

$$f_1'(x) = e^x + xe^x$$

Differentiating again gives us:

$$f_1''(x) = 2e^x + xe^x$$

Following this clear pattern allows us to determine that:

$$f_1^n(x) = ne^x + xe^x$$

Thus,

$$f_1^n(0) = n$$

This result is very useful, having many applications such as being used to solve non-homogeneous recurrence relations. Now consider the following function:

$$f_2(x) = (x + x^2) e^x$$

This function also constitutes as a polynomial-exponential product function whose polynomial factor has a degree of 2. Differentiating this function using the product rule gives us:

$$f_2'(x) = (1 + 3x + x^2) e^x$$

Differentiating again shows that,

$$f_2''(x) = (4 + 5x + x^2) e^x$$

The pattern of the n -th derivative is harder to spot here. As f_2 is differentiated n times, $C(f_2^n, 2) = 1$. Because of this, when f_2^n is differentiated, 2 (1 multiplied by the 2 of is added to the coefficient of the x term. This implies that $C(f_2^n, 1) = 2n + 1$, as 2 is added n times to 1

(the initial value of the coefficient). It follows that, when f_2^n is differentiated, $2n + 1$ is added to the constant term. This gives us the following expression:

$$C(f_2^n, 0) = \sum_{i=0}^{n-1} 2i + 1 = 2 \sum_{i=0}^{n-1} i + \sum_{i=0}^{n-1} 1$$

Applying standard summation formulae let's us determine that,

$$\begin{aligned} C(f_2^n, 0) &= n(n-1) + n = n^2 - n + n \\ \therefore C(f_2^n, 0) &= n^2 \end{aligned}$$

Overall, we can determine the following expression:

$$f_2^n(x) + (n^2 + (2n+1)x + x^2) e^x,$$

Thus,

$$f_2^n(0) = n^2$$

Now let's consider the following polynomial-exponential product function, whose polynomial factor has a degree of 3, which appears to follow the same pattern as the previous functions considered:

$$f_3(x) = (x + x^2 + x^3)e^x$$

$C(f_3^n, 3) = 1$. Because of this, when f_3 is differentiated n times, 3 is added to the coefficient of x^2 n times. Therefore, we can determine that $C(f_3^n, 2) = 3n + 1$. Thus,

$$C(f_3^n, 1) = 1 + 2 \sum_{i=0}^{n-1} 3i + 1 = 1 + 6 \sum_{i=0}^{n-1} i + 2 \sum_{i=0}^{n-1} 1$$

Applying standard summation formulae again gives us:

$$\begin{aligned} C(f_3^n, 1) &= 1 + 3(n(n-1)) + 2n = 1 + 3n^2 - 3n + 2n \\ \therefore C(f_3^n, 1) &= 3n^2 - n + 1 \end{aligned}$$

Thus,

$$C(f_3^n, 0) = \sum_{i=0}^{n-1} 3i^2 - i + 1 = 3 \sum_{i=0}^{n-1} i^2 - \sum_{i=0}^{n-1} i + \sum_{i=0}^{n-1} 1$$

Using standard summation formula once again, gives us:

$$\begin{aligned} C(f_3^n, 0) &= \frac{n(n-1)(2n-1)}{2} - \frac{n(n-1)}{2} + n = \frac{2n^3 - 4n^2 + 2n}{2} + n \\ \therefore C(f_3^n, 0) &= n^3 - 2n^2 + 2n \end{aligned}$$

Overall, we can determine the expression:

$$f_3^n(x) = ((n^3 - 2n^2 + 2n) + (3n^2 - n + 1)x + (3n + 1)x^2 + x^3) e^x$$

Thus,

$$f_3^n(0) = n^3 - 2n^2 + 2n$$

Therefore, $f_3^n(0) \neq n^q$. Because it doesn't satisfy the equation, this is not a function we are looking for. Instead, let's reassign f_3 :

$$f_3(x) = (x + 3x^2 + x^3) e^x$$

$$C(f_3^n, 3) = 1 \implies C(f_3^n, 2) = 3n + 3. \text{ Thus,}$$

$$\begin{aligned} C(f_3^n, 1) &= 1 + 2 \sum_{i=0}^{n-1} 3i + 3 = 1 + 6 \sum_{i=0}^{n-1} i + 6 \sum_{i=0}^{n-1} 1 \\ \therefore C(f_3^n, 1) &= 3n^2 + 3n + 1 \end{aligned}$$

Hence,

$$\begin{aligned} C(f_3^n, 0) &= \sum_{i=0}^{n-1} 3i^2 + 3i + 1 = 3 \sum_{i=0}^{n-1} i^2 + 3 \sum_{i=0}^{n-1} i + \sum_{i=0}^{n-1} 1 \\ \therefore C(f_3^n, 0) &= n^3 \end{aligned}$$

So we can deduce the following:

$$\begin{aligned} f_3^n(x) &= (n^3 + (3n^2 + 3n + 1)x + (3n + 3)x^2 + x^3) e^x \\ \therefore f_3^n(0) &= n^3 \end{aligned}$$

This is a function we are looking for. These 3 functions discovered all satisfy $f^n(0) = n^q$ with different q . As these 3 functions are all polynomial-exponential product functions, it is reasonable to assume that all functions within the family of analytic functions which satisfy $f^n(0) = n^q$ ($q \geq 0$, $n, q \in \mathbb{Z}$) will be polynomial-exponential product functions of some form. So far, this is not proven to be the case, however, this deduction serves as reason to focus the investigation on these functions.

1.2.2 Definition

It is reasonable to conjecture that, for a polynomial-exponential product function to satisfy $f^n(0) = n^p$, it is necessary to have a polynomial factor of degree p , as this has been the case so far. It is well known that

$$\sum_{i=0}^n a_q i^q + a_{q-1} i^{q-1} + \dots + a_1 i + a_0$$

is a polynomial in n of degree $q + 1$. For a polynomial exponential function,

$$f(x) = (\lambda_0 + \lambda_1 x + \dots + \lambda_p x^p) e^x,$$

it is clear that $C(f^n, p) = \lambda_p$, because the coefficient of x^p remains unchanged as f is differentiated. Each time f is differentiated, p multiplied by the coefficient of x^p is added to the coefficient of x^{p-1} , which has the initial (when f is undifferentiated) value of $C(f, p-1)$. Hence,

$$C(f^n, p-1) = C(f, p-1) + p \sum_{i=0}^{n-1} \lambda_p = C(f, p-1) + p \lambda_p \sum_{i=0}^{n-1} i^0$$

$C(f, p-1)$, p , and λ_p are constant. Therefore, as a result of the summation in the expression of $C(f^n, p-1)$, we can determine that $C(f^n, p-1)$ is a polynomial in n of degree 1. Specifically,

$$C(f^n, p-1) = p \lambda_p n + C(f, p-1)$$

When f^{n-1} is differentiated, $(p-1) \cdot C(f^{n-1}, p-1)$ is added to the coefficient of x^{p-2} . Thus,

$$C(f^n, p-2) = C(f, p-2) + (p-1) \cdot \sum_{i=0}^{n-1} C(f^i, p-1)$$

As $C(f^i, p-1)$ is a polynomial in i of degree 1 and $C(f, p-2)$ and $(p-1)$ are simply constants, we can deduce that $C(f^n, p-2)$ is a polynomial in n of degree 2. These two instances enable us to determine the pattern for evaluating $C(f^n, p-r)$, which is as follows:

$$C(f^n, p-r) = C(f, p-r) + (p-r+1) \cdot \sum_{i=0}^{n-1} C(f^i, p-r+1)$$

Thus, the degree of $C(f^n, p-r)$ is one greater than the degree of $C(f^n, p-r+1)$. This has been shown in these instances where $C(f^n, p)$ is a polynomial in n of degree 0, so that $C(f^n, p-1)$ is of degree 1, and that $C(f^n, p-2)$ is of degree 2. Thus, as $C(f^n, p)$ is always a polynomial in n of degree 0 (as it is constant), we can deduce that $C(f^n, p-r)$ will be a polynomial in n of degree r . So we have shown that $C(f^n, p-r)$ is a polynomial in n of degree r .

$f^n(0) = C(f^n, 0) = C(f^n, p-p) \implies r = p$. Therefore, $f^n(0)$ will be a polynomial in n of degree p . We are trying to find functions which satisfy $f^n(0) = n^q$, which is a polynomial in n of degree q . Therefore, as $f^n(0)$ is of degree p , to satisfy $f^n(0) = n^q$ we require $q = p$. So essentially, $f^n(0) = n^q$ can only hold when $f^n(0) = n^p$. Therefore proving our conjecture that, for a polynomial-exponential product function to satisfy $f^n(0) = n^p$, it is necessary to have a polynomial factor of degree p .

When discovering the family of polynomial-exponential product functions which satisfy $f^n(0) = n^p$, no obvious pattern was apparent in the values of the coefficients of the polynomial factors. However, since some of these coefficients take different values for different values of p , they must depend on p .

Overall, based on deduction and results, we can define our family of polynomial-exponential product functions which satisfy $f^n(0) = n^p$ to be of the following form:

$$f_p(x) = (\lambda_0(p) + \lambda_1(p)x + \cdots + \lambda_p(p)x^p) e^x$$

Where:

- $\lambda_0, \lambda_1, \dots, \lambda_p$ are functions of p , each returning a constant coefficient,
- p is the degree of the polynomial factor.

Suppose now that want to extend this family of functions to functions which satisfy $f^n(0) = (k+n)^p$, which makes these functions more useful, so that they can be applied to problems such as non-homogeneous recurrence relations more flexibly. If $f_p^n(0) = n^p$, so is a member of our current family, then $f_p^k(0) = k^p$, where k is a non-negative integer. Therefore, $f_p^{k+n}(0) = (k+n)^p$. So, if we can find these functions satisfying $f_p^n(0) = n^p$, we can simply differentiate these k times to derive our function f_p^k , so that $f_p^{k+n}(0) = (k+n)^p$. Following this, I will now define our family of polynomial-exponential product functions satisfying $f^n(0) = (k+n)^p$ as having the following form:

$$f_p^k(x) = (\lambda_0(p, k) + \lambda_1(p, k)x + \cdots + \lambda_p(p, k)x^p) e^x$$

Where:

- $\lambda_0, \lambda_1, \dots, \lambda_p$ are functions of p and k , each returning a constant coefficient,
- p is the degree of the polynomial factor.
- k is a non-negative integer, representing the number of times f_p has been initially differentiated, so that instead of $f_p^n(0) = n^p$, we have $f_p^{k+n}(0) = (k+n)^p$.

I have made these coefficient functions, λ_r , depend on k as well as p , because they will generally change as f_p is differentiated k times.

2 Structure derivation

2.1 Structure of coefficients of polynomial-exponential product functions

The proof presented in this subsection is included for completeness and is not required for understanding the remainder of the paper. Readers primarily interested in the final result should skip to the concluding formula without affecting their understanding of subsequent sections.

Following our definitions, we will let f_p^k denote our polynomial-exponential product function which satisfies $f_p^{k+n}(0) = (k+n)^p$ where $k, n, p \geq 0$ and $k, n, p \in \mathbb{Z}$. This has the following form:

$$f_p^k(x) = (\lambda_0(p, k) + \lambda_1(p, k)x + \cdots + \lambda_p(p, k)x^p)e^x$$

Throughout the derivation, r will be treated as representing the r -th term in the polynomial factor of these functions. For example, $C(f_p^k, r)$ will be the coefficient of the x^r term in the polynomial factor of f_p^k . It follows that $0 \leq r \leq p$ and $r \in \mathbb{Z}$.

When f_p^{k+n-1} is differentiated, the coefficient of the x^r term becomes the coefficient of the x^r term of f_p^{k+n-1} added to the coefficient of the x^{r+1} term of f_p^{k+n-1} multiplied by $r+1$. Thus,

$$C(f_p^{k+n}, r) = C(f_p^{k+n-1}, r) + (r+1) \cdot C(f_p^{k+n-1}, r+1)$$

We can use this equation to determine useful results.

$$C(f_p^k, r) = \lambda_r(p, k) \implies C(f_p^{k+1}, r) = \lambda_r(p, k) + (r+1) \cdot \lambda_{r+1}(p, k)$$

Thus,

$$C(f_p^{k+1}, r+1) = \lambda_{r+1}(p, k) + (r+2) \cdot \lambda_{r+2}(p, k)$$

$$\therefore C(f_p^{k+2}, r) = \lambda_r(p, k) + 2(r+1) \cdot \lambda_{r+1}(p, k) + (r+1)(r+2) \cdot \lambda_{r+2}(p, k)$$

Similarly,

$$C(f_p^{k+2}, r+1) = \lambda_{r+1}(p, k) + 2(r+2) \cdot \lambda_{r+2}(p, k) + (r+2)(r+3) \cdot \lambda_{r+3}(p, k)$$

Therefore,

$$C(f_p^{k+3}, r) = \lambda_r(p, k) + 3(r+1) \cdot \lambda_{r+1}(p, k) + 3(r+1)(r+2) \cdot \lambda_{r+2}(p, k) + (r+1)(r+2)(r+3) \cdot \lambda_{r+3}(p, k)$$

It is clear that each $\lambda_{r+i}(p, k)$ term of $C(f_p^{k+n}, r)$ is multiplied by $(r+1)(r+2) \dots (r+i)$. This happens through differentiation. For example, if consider the following function:

$$f_2^k(x) = (\lambda_0(2, k) + \lambda_1(2, k)x + \lambda_2(2, k)x^2) e^x$$

Then,

$$f_2^{k+1} = ((\lambda_0(2, k) + \lambda_1(2, k)) + (\lambda_1(2, k) + 2\lambda_2(2, k))x + \lambda_2(2, k)x^2) e^x$$

Therefore,

$$f_2^{k+2} = ((\lambda_0(2, k) + 2\lambda_1(2, k) + 2\lambda_2(2, k)) + \cdots) e^x$$

The $\lambda_2(p, k)$ term in $C(f_p^{k+2}, 0)$ is multiplied by $2 \cdot 1 = (0+1)(0+2)$, because it has been multiplied by the 2 of x^2 and then multiplied by the 1 of x^1 . Therefore, all terms, $\lambda_{r+i}(p, k)$, in $C(f_p^{k+n}, r)$ will be multiplied by $(r+1)(r+2) \dots (r+i) = \frac{(r+i)!}{r!}$.

Let's now prove something simple indirectly related to our working. Consider the following series:

$$\sum_{i=0}^n \binom{n}{i} a_i = \binom{n}{0} a_0 + \binom{n}{1} a_1 + \cdots + \binom{n}{n-1} a_{n-1} + \binom{n}{n} a_n$$

Now, consider the similar series:

$$\sum_{i=0}^n \binom{n}{i} a_{i+1} = \binom{n}{0} a_1 + \binom{n}{1} a_2 + \cdots + \binom{n}{n-1} a_n + \binom{n}{n} a_{n+1}$$

For both of these series, a_i represents a general term in the series. n represents the upper limit and is a non-negative integer. For these 2 distinct series, any term of a_i in 1 series is equal to the a_i in the other series. Now, add these series together:

$$\sum_{i=0}^n \binom{n}{i} a_i + \sum_{i=0}^n \binom{n}{i} a_{i+1} = \binom{n}{0} a_0 + \left[\binom{n}{1} + \binom{n}{0} \right] a_1 + \cdots + \left[\binom{n}{n} + \binom{n}{n-1} \right] a_n + \binom{n}{n} a_{n+1}$$

When we have,

$$\binom{n}{i} + \binom{n}{i-1},$$

we can determine that this equal to

$$\frac{n!}{i!(n-i)!} + \frac{n!}{(i-1)!(n-i+1)!} = \frac{n!(n-i+1) + n!(i)}{i!(n+1-i)!} = \frac{n!(n+1)}{i!(n+1-i)!} = \frac{(n+1)!}{i!(n+1-i)!} = \binom{n+1}{i}.$$

Therefore,

$$\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}.$$

This result allows us to deduce that,

$$\sum_{i=0}^n \binom{n}{i} a_i + \sum_{i=0}^n \binom{n}{i} a_{i+1} = \binom{n}{0} a_0 + \binom{n+1}{1} a_1 + \cdots + \binom{n+1}{n} a_n + \binom{n}{n} a_{n+1}.$$

Furthermore, $\binom{n}{0} = \binom{n+1}{0} = 1$ and $\binom{n}{n} = \binom{n+1}{n+1} = 1$. This allows us to further deduce that,

$$\sum_{i=0}^n \binom{n}{i} a_i + \sum_{i=0}^n \binom{n}{i} a_{i+1} = \binom{n+1}{0} a_0 + \binom{n+1}{1} a_1 + \cdots + \binom{n+1}{n} a_n + \binom{n+1}{n+1} a_{n+1}.$$

Therefore, we have proven the following result:

$$\sum_{i=0}^n \binom{n}{i} a_i + \sum_{i=0}^n \binom{n}{i} a_{i+1} = \sum_{i=0}^{n+1} \binom{n+1}{i} a_i \quad (1)$$

This Now let's prove a similar simple result. Consider the following 2 series:

$$\sum_{i=0}^p \binom{n}{i} a_i = \binom{n}{0} a_0 + \binom{n}{1} a_1 + \cdots + \binom{n}{p-1} a_{p-1} + \binom{n}{p} a_p,$$

$$\sum_{i=0}^{p-1} \binom{n}{i} a_{i+1} = \binom{n}{0} a_1 + \binom{n}{1} a_2 + \cdots + \binom{n}{p-2} a_{p-1} + \binom{n}{p-1} a_p.$$

Where p is a non-negative integer and $p < n$. These series are very similar to the previous series considered, sharing the same properties apart from having differing upper limits. We can deduce that,

$$\sum_{i=0}^p \binom{n}{i} a_i + \sum_{i=0}^{p-1} \binom{n}{i} a_{i+1} = \binom{n}{0} a_0 + \left[\binom{n}{1} + \binom{n}{0} \right] a_1 + \cdots + \left[\binom{n}{p} + \binom{n}{p-1} \right] a_p.$$

Therefore,

$$\sum_{i=0}^p \binom{n}{i} a_i + \sum_{i=0}^{p-1} \binom{n}{i} a_{i+1} = \binom{n+1}{0} a_0 + \binom{n+1}{1} a_1 + \cdots + \binom{n+1}{p-1} a_p + \binom{n+1}{p} a_p.$$

Therefore, we have proven the following result:

$$\sum_{i=0}^p \binom{n}{i} a_i + \sum_{i=0}^{p-1} \binom{n}{i} a_{i+1} = \sum_{i=0}^p \binom{n+1}{i} a_i \quad (2)$$

These results will be useful later. For now, consider an arbitrary polynomial-exponential product function,

$$f_p^k(x) = (\lambda_0(p, k) + \lambda_1(p, k)x + \cdots + \lambda_{p-2}(p, k)x^{p-2} + \lambda_{p-1}(p, k)x^{p-1} + \lambda_p(p, k)x^p) e^x.$$

Differentiating gives us the following:

$$f_p^{k+1}(x) = (\cdots + (\lambda_{p-2}(p, k) + (p-1)\lambda_{p-1}(p, k))x^{p-2} + (\lambda_{p-1}(p, k) + p\lambda_p(p, k))x^{p-1} + \lambda_p x^p) e^x$$

It is clear that $C(f_p^{k+1}, r) = \lambda_r(p, k) + (r+1)\lambda_{r+1}(p, k)$ for all terms apart from $C(f_p^{k+1}, p)$. Now, if we differentiate again, we get the following:

$$f_p^{k+2}(x) = (\cdots + (\lambda_{p-2} + 2(p-1)\lambda_{p-1} + (p-1)(p)\lambda_p)x^{p-2} + (\lambda_{p-1} + 2p\lambda_p)x^{p-1} + \lambda_p x^p) e^x$$

λ_i is simply a shorthand expression of $\lambda_i(p, k)$ here and should be interpreted as so hereafter for the remainder of this subsection. Here we can see that $C(f_p^{k+2}, r) = \lambda_r + 2(r+1)\lambda_{r+1} + (r+1)(r+2)\lambda_{r+2}$ apart from the last 2 terms. This is because $C(f_p^{k+2}, p-1)$ cannot have a λ_{r+2} term, as this would represent λ_{p+1} , and x^{p+1} is not a term in the initial polynomial. However, we can see it still follows the same general pattern as $C(f_p^{k+2}, r)$ for $r \leq p-2$, but instead cuts off the series when the upper limit of p is reached. So we can determine that these series are not cut off where $r \leq p-n$. It appears that the pattern of the $C(f_p^{k+n}, r)$ series follows the binomial distribution, so that each λ_i term is multiplied by $\binom{n}{i}$.

$$C(f_p^k, r) = \lambda_r$$

Therefore,

$$C(f_p^k, r) = \sum_{i=0}^0 \binom{0}{i} \frac{(r+i)!}{r!} \lambda_{r+i}$$

Is valid. We have already proven the purpose of $\frac{(r+1)!}{r!}$, so has been included implicitly. So, for $r \leq p-1$:

$$C(f_p^{k+1}, r) = \sum_{i=0}^0 \binom{0}{i} \frac{(r+i)!}{r!} \lambda_{r+i} + (r+1) \sum_{i=0}^0 \binom{0}{i} \frac{(r+1+i)!}{(r+1)!} \lambda_{r+1+i}$$

Therefore, using result (1), we get the following:

$$C(f_p^{k+1}, r) = \sum_{i=0}^1 \binom{1}{i} \frac{(r+i)!}{r!} \lambda_{r+i}$$

Therefore, for $r \leq p-2$:

$$C(f_p^{k+2}, r) = \sum_{i=0}^1 \binom{1}{i} \frac{(r+i)!}{r!} \lambda_{r+i} + (r+1) \sum_{i=0}^1 \binom{1}{i} \frac{(r+1+i)!}{(r+1)!} \lambda_{r+1+i}$$

So,

$$C(f_p^{k+2}, r) = \sum_{i=0}^2 \binom{2}{i} \frac{(r+i)!}{r!} \lambda_{r+i}.$$

Clearly, this pattern has to continue for all r where $r \leq p-n$, where n represents how many times f_p^k has been differentiated. Therefore, we have proven the following result:

$$C(f_p^{k+n}, r) = \sum_{i=0}^n \binom{n}{i} \frac{(r+i)!}{r!} \lambda_{r+i}$$

When $r \leq p-n$. However, if $r > p-n$, then the series will simply cut off at $i = p$, as observed. We can make this more rigorous using result (2). We can easily obtain the following:

$$C(f_p^{k+1}, p) = \lambda_p = \sum_{i=0}^0 \binom{0}{i} \frac{(p+i)!}{p!} \lambda_{p+i}$$

And,

$$C(f_p^{k+1}, p-1) = \sum_{i=0}^1 \binom{1}{i} \frac{(p-1+i)!}{(p-1)!} \lambda_{p-1+i}.$$

$$C(f_p^{k+2}, p-1) = C(f_p^{k+1}, p-1) + pC(f_p^{k+1}, p)$$

Therefore, using result (2) shows that:

$$C(f_p^{k+2}, p-1) = \sum_{i=0}^1 \binom{2}{i} \frac{(p-1+i)!}{(p-1)!} \lambda_{p-1+i}$$

Further, we can deduce the following:

$$C(f_p^{k+3}, p-1) = \sum_{i=0}^1 \binom{3}{i} \frac{(p-1+i)!}{(p-1)!} \lambda_{p-1+i}$$

So it is shown that the result for $C(f_p^{k+n}, r)$ holds only with a different upper limit of the summation, which is $p-r$, so that this series does not exceed the term of λ_p .

Now, I will rewrite the final formula we have derived without any shorthand:

$$C(f_p^{k+n}, r) = \sum_{i=0}^{\min(n, p-r)} \binom{n}{i} \frac{(r+i)!}{r!} \lambda_{r+i}(p, k)$$

Where:

- $k \geq 0, k \in \mathbb{Z}$
- $n \geq 0, n \in \mathbb{Z}$
- $p \geq 0, p \in \mathbb{Z}$
- $0 \leq r \leq p, r \in \mathbb{Z}$

2.2 General structure of the family

Using the final result from the previous subsection, we can determine the following result:

$$C(f_p^{k+n}, 0) = \sum_{i=0}^{\min(n,p)} \binom{n}{i} \frac{(i)!}{0!} \lambda_i(p, k) = \sum_{i=0}^{\min(n,p)} \binom{n}{i} (i!) \lambda_i(p, k)$$

For now, suppose $n \leq p$. Therefore,

$$C(f_p^{k+n}, 0) = \sum_{i=0}^n \binom{n}{i} (i!) \lambda_i(p, k).$$

$C(f_p^{k+n}, 0) = f_p^{k+n}(0)$. As we stated previously for f_p^k to be a member of the family of functions we are investigating, it must satisfy $f_p^{k+n}(0) = (k+n)^p$. So for f_p^k to be a member of the family, the following must be true:

$$(k+n)^p = C(f_p^{k+n}, 0) \implies (k+n)^p = \sum_{i=0}^n \binom{n}{i} (i!) \lambda_i(p, k)$$

Taking the last term out of the summation series and rearranging gives us the following expression:

$$n! \lambda_n(p, k) = (k+n)^p - \sum_{i=0}^{n-1} \binom{n}{i} (i!) \lambda_i(p, k) \quad (1)$$

Taking the next last term out of the summation and further rearranging gives us:

$$n! \lambda_n(p, k) = (k+n)^p - \binom{n}{n-1} (n-1)! \lambda_{n-1}(p, k) - \sum_{i=0}^{n-2} \binom{n}{i} (i!) \lambda_i(p, k)$$

Using (1), we can determine the following:

$$(n-1)! \lambda_{n-1}(p, k) = (k+n-1)^p - \sum_{i=0}^{n-2} \binom{n-1}{i} (i!) \lambda_i(p, k)$$

Substituting this obtains:

$$n! \lambda_n(p, k) = (k+n)^p - \binom{n}{n-1} \left[(k+n-1)^p - \sum_{i=0}^{n-2} \binom{n-1}{i} (i!) \lambda_i(p, k) \right] - \sum_{i=0}^{n-2} \binom{n}{i} (i!) \lambda_i(p, k)$$

Therefore, giving the result:

$$n! \lambda_n(p, k) = (k+n)^p - \binom{n}{n-1} (k+n-1)^p + \binom{n}{n-1} \sum_{i=0}^{n-2} \binom{n-1}{i} (i!) \lambda_i(p, k) - \sum_{i=0}^{n-2} \binom{n}{i} (i!) \lambda_i(p, k)$$

Now, let's take out the last term of both summation series present in the current equation:

$$\begin{aligned} n!\lambda_n(p, k) &= (k+n)^p - \binom{n}{n-1}(k+n-1)^p + \binom{n}{n-1}\binom{n-1}{n-2}(n-2)!\lambda_{n-2}(p, k) \\ &\quad - \binom{n}{n-2}(n-2)!\lambda_{n-2}(p, k) + \binom{n}{n-1}\sum_{i=0}^{n-3}\binom{n-1}{i}(i!)\lambda_i(p, k) - \sum_{i=0}^{n-3}\binom{n}{i}(i!)\lambda_i(p, k) \end{aligned}$$

Using (1) again, allows us to obtain the following:

$$\begin{aligned} n!\lambda_n(p, k) &= (k+n)^p - \binom{n}{n-1}(k+n-1)^p + \left[\binom{n}{n-1}\binom{n-1}{n-2} - \binom{n}{n-2} \right] \\ &\quad \times \left[(k+n-2)^p - \sum_{i=0}^{n-3}\binom{n-2}{i}(i!)\lambda_i(p, k) \right] + \binom{n}{n-1}\sum_{i=0}^{n-3}\binom{n-1}{i}(i!)\lambda_i(p, k) - \sum_{i=0}^{n-3}\binom{n}{i}(i!)\lambda_i(p, k). \end{aligned}$$

Now consider the following result:

$$\binom{n}{n-1}\binom{n-1}{n-2} - \binom{n}{n-2} = \frac{n!}{(n-2)!} - \frac{n!}{2!(n-2)!} = \frac{n!}{2!(n-2)!} = \binom{n}{n-2}$$

Applying this to our expression gives us:

$$\begin{aligned} n!\lambda_n(p, k) &= (k+n)^p - \binom{n}{n-1}(k+n-1)^p + \binom{n}{n-2}(k+n-2)^p \\ &\quad - \binom{n}{n-2}\sum_{i=0}^{n-3}\binom{n-2}{i}(i!)\lambda_i(p, k) + \binom{n}{n-1}\sum_{i=0}^{n-3}\binom{n-1}{i}(i!)\lambda_i(p, k) - \sum_{i=0}^{n-3}\binom{n}{i}(i!)\lambda_i(p, k) \end{aligned}$$

So far, each time the last term is taken out of all summation series in $n!\lambda_n(p, k)$, we can take out $(n-q)!\lambda_{n-q}(p, k)$ from all of these terms, which has been multiplied by the following series:

$$\sum_{i=0}^{q-1}(-1)^{i+1}\binom{n}{n-i}\binom{n-i}{n-q} = \sum_{i=0}^{q-1}(-1)^{i+1}\frac{n!}{i!(n-q)!(q-i)!} = \frac{n!}{(n-q)!}\sum_{i=0}^{q-1}(-1)^{i+1}\frac{1}{i!(q-i)!}$$

$q \geq 0, q \in \mathbb{Z}$. Let

$$S = \sum_{i=0}^{q-1}(-1)^{i+1}\frac{1}{i!(q-i)!}$$

hereafter. When q is odd:

$$S = -\frac{1}{0!(q-0)!} + \frac{1}{1!(q-1)!} - \frac{1}{2!(q-2)!} \cdots + \frac{1}{(q-2)!2!} - \frac{1}{(q-1)!1!}$$

So, when q is odd:

$$S = -\frac{1}{q!}$$

Now, using the partial fractions method, consider the following:

$$\frac{1}{q!} = \frac{a_0}{q} + \frac{a_1}{q-1} + \cdots + \frac{a_{q-2}}{2} + \frac{a_{q-1}}{1} = \frac{a_0}{q} + \frac{a_1}{q-1} + \cdots + \frac{a_{q-2}}{q-(q-2)} + \frac{a_{q-1}}{q-(q-1)}$$

Where a_i are constant. Therefore,

$$1 = a_0(q-1)! + a_1q(q-2)! + \cdots + a_{q-2}q(q-1) \cdots (q-(q-3))(q-(q-1)) + a_{q-1}q(q-1) \cdots (q-(q-3))(q-(q-2)).$$

For a_i , if we let $q = i$, as all other terms are multiplied by $(q - i)$, thereby evaluating to 0. Consequently, we can determine the following result:

$$1 = a_i(i)(i-1) \dots (i-i+1)(i-i-1) \dots (i-(q-1)) = a_i(i!)(-1) \dots (-(q-i-1))$$

The q remaining here represent the constant value of q , so have not been set to equal i . If q is even, then a_i will be multiplied by an odd number of terms. For example, a_0 is multiplied by $(q-1)!$, which will be composed of $q-1$ terms; q is even implying $q-1$ is odd. Now, if i is even, so $i!$ has an even number of terms, then it follows that $(-1) \dots (-(q-i-1))$ would have an odd number of terms so that a_i is still multiplied by an odd number of terms (given q is even). If $(-1) \dots (-(q-i-1))$ has an odd number of terms, then this will evaluate to $-(q-i-1)!$. Otherwise, if i is odd, then $(-1) \dots (-(q-i-1))$ would have an even number of terms, thereby evaluating to $(q-i-1)!$. Therefore, giving us the following result:

$$1 = a_i(i!)(-1)^{i+1}(q-i-1)! \implies a_i = \frac{1}{(i!)(-1)^{i+1}(q-i-1)!}$$

Therefore, when q is even, we get the following:

$$\frac{1}{q!} = \sum_{i=0}^{q-1} \frac{a_i}{q-i} = \sum_{i=0}^{q-1} \frac{1}{(i!)(-1)^{i+1}(q-i)!} = S$$

Therefore:

$$S = \frac{1}{q!} \text{ when } q \text{ is even}$$

$$S = -\frac{1}{q!} \text{ when } q \text{ is odd}$$

Now that we have determined the value of S when q is even or odd, we can determine the following result:

$$\frac{n!}{(n-q)!} \sum_{i=0}^{q-1} (-1)^{i+1} \frac{1}{i!(q-i)!} = \frac{n!}{(n-q)!} S = (-1)^q \frac{n!}{q!(n-q)!} = (-1)^q \binom{n}{n-q} \quad (2)$$

Therefore, when the last term is taken out of all of the summation series in $n!\lambda_n(p, k)$, and we take out $(n-q)!\lambda_{n-q}(p, k)$ from this term (after collecting like terms), it is multiplied by this result (2). Therefore, applying results (1) and (2), we can deduce that this term will have the following form:

$$(-1)^q \binom{n}{n-q} \left((k+n-q)^p - \sum_{i=0}^{n-q-1} \binom{n-q}{i} (i!) \lambda_i(p, k) \right)$$

Thereby adding the following terms to the series of $n!\lambda_n(p, k)$:

$$(-1)^q \binom{n}{n-q} (k+n-q)^p - (-1)^q \binom{n}{n-q} \sum_{i=0}^{n-q-1} \binom{n-q}{i} (i!) \lambda_i(p, k)$$

Therefore, we can determine that the following series,

$$\begin{aligned} n!\lambda_n(p, k) &= (k+n)^p - \binom{n}{n-1} (k+n-1)^p + \binom{n}{n-2} (k+n-2)^p \\ &- \binom{n}{n-2} \sum_{i=0}^{n-3} \binom{n-2}{i} (i!) \lambda_i(p, k) + \binom{n}{n-1} \sum_{i=0}^{n-3} \binom{n-1}{i} (i!) \lambda_i(p, k) - \sum_{i=0}^{n-3} \binom{n}{i} (i!) \lambda_i(p, k), \end{aligned}$$

Will further expand into the following:

$$\begin{aligned}
n!\lambda_n(p, k) &= (k+n)^p - \binom{n}{n-1}(k+n-1)^p + \binom{n}{n-2}(k+n-2)^p \\
&\quad - \binom{n}{n-3}(k+n-3)^p + \binom{n}{n-3} \sum_{i=0}^{n-4} \binom{n-3}{i} (i!) \lambda_i(p, k) \\
&\quad - \binom{n}{n-2} \sum_{i=0}^{n-4} \binom{n-2}{i} (i!) \lambda_i(p, k) + \binom{n}{n-1} \sum_{i=0}^{n-4} \binom{n-1}{i} (i!) \lambda_i(p, k) - \sum_{i=0}^{n-4} \binom{n}{i} (i!) \lambda_i(p, k),
\end{aligned}$$

It follows that the next last term, of $(n-4)\lambda_{n-4}(p, k)$, taken out of all of these series will be again multiplied by the following series:

$$\sum_{i=0}^{q-1} (-1)^{i+1} \binom{n}{n-i} \binom{n-i}{n-q} = (-1)^q \binom{n}{n-q}$$

Thus, as it is shown to continue the observed pattern, we can observe the following:

$$\begin{aligned}
n!\lambda_n(p, k) &= \binom{n}{n}(k+n)^p - \binom{n}{n-1}(k+n-1)^p + \cdots + (-1)^i \binom{n}{n-i}(k+n-i)^p + \cdots \\
&\quad + (-1)^{n-1} \binom{n}{1}(k+1)^p - \binom{n}{n} \sum_{i=0}^0 \binom{n}{i} (i!) \lambda_i(p, k) + \binom{n}{n-1} \sum_{i=0}^0 \binom{n}{i} (i!) \lambda_i(p, k) + \cdots \\
&\quad + (-1)^{n-1} \binom{n}{1} \sum_{i=0}^0 \binom{n}{i} (i!) \lambda_i(p, k)
\end{aligned}$$

Again, using (2), we can deduce that this is equal to the following:

$$\begin{aligned}
n!\lambda_n(p, k) &= \binom{n}{n}(k+n)^p - \binom{n}{n-1}(k+n-1)^p + \cdots + (-1)^i \binom{n}{n-i}(k+n-i)^p + \cdots \\
&\quad + (-1)^{n-1} \binom{n}{1}(k+1)^p + (-1)^n \binom{n}{0} \lambda_0(p, k)
\end{aligned}$$

$C(f_p^k, 0) = \lambda_0(p, k)$, so it is necessary that, $\lambda_0(p, k) = (k)^p$. Therefore, we can determine the following:

$$\lambda_n(p, k) = \frac{\binom{n}{n}(k+n)^p - \binom{n}{n-1}(k+n-1)^p + \cdots + (-1)^{n-1} \binom{n}{1}(k+1)^p + (-1)^n \binom{n}{0}(k)^p}{n!}$$

Overall, we can conclude the following result:

$$\lambda_n(p, k) = \frac{1}{n!} \sum_{i=0}^n (-1)^i \binom{n}{n-i} (k+n-i)^p$$

Let's reassign variables:

$$\lambda_r(p, k) = \frac{1}{r!} \sum_{i=0}^r (-1)^i \binom{r}{r-i} (k+r-i)^p$$

Now, this only holds if $r \leq p$, as we assumed $r = n \leq p$ at the beginning of this subsection. However, because this is an equation for the coefficient of the x^r term in the polynomial factor

of a polynomial-exponential product function with degree p , r must be less than or equal to p . Therefore, this result holds for all r that we can have, as $r \leq p$ is necessary.

Ultimately, this allows us to derive a general form equation for all of members of the family of polynomial-exponential product functions, f_p^k , which satisfy $f_p^{k+n}(0) = (k+n)^p$:

$$f_p^k(x) = e^x \sum_{r=0}^p \left(\frac{x^r}{r!} \sum_{i=0}^r (-1)^i \binom{r}{r-i} (k+r-i)^p \right)$$

Where:

- $x \in \mathbb{R}$,
- $k, n, p \geq 0, k, n, p \in \mathbb{Z}$.